Assignment 10: Event Financial Planning

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Calculate the overall operational cost of hosting a campus tech fest. Offset costs using sponsorship amounts, then determine the mathematical minimum number of registrations required to break even using purely integer operations.

Algorithm

1. Define a tuple packing categories and their respective costs sequentially (alternating strings and ints).
2. Iterate through the tuple using a loop with step indexing to sum only the numerical cost values.
3. Calculate the shortfall by subtracting the total sponsorship from the aggregated total event cost.
4. Calculate the minimum required attendees using floor division logic to emulate mathematical ceiling: `(shortfall + fee - 1) // fee`.
5. Print the financial breakdown.

Python Implementation

```
# Tuple containing (Category, Cost, Category, Cost...)
event_data = (
    "Venue", 25000,
    "Equipment", 8000,
    "Catering", 30000,
    "Certificates", 2500,
    "Transport", 4500
)

total_cost = 0
# Extract numerical costs by looping over odd indices
for i in range(1, len(event_data), 2):
    total_cost += event_data[i]

sponsorship = 40000
registration_fee = 600

shortfall = total_cost - sponsorship

# Finding minimum registrations to cover the shortfall
# Using (A + B - 1) // B trick to perform integer ceiling division
min_registrations = 0
if shortfall > 0:
    min_registrations = (shortfall + registration_fee - 1) // registration_fee

print("--- Tech Fest Financials ---")
print("Total Expected Cost    : Rs.", total_cost)
print("Sponsorship Available  : Rs.", sponsorship)
print("Budget Shortfall       : Rs.", shortfall)
print("Registration Fee       : Rs.", registration_fee)
print("Min. Registrations Req.:", min_registrations, "students")
```